

Soldering Techniques

Hand • Hot Air • Reflow — from kits to production

The skill that turns a schematic into a working board

Audience: Beginner → Intermediate Engineering Students

Theory • Flux • Iron • Hot-Air • Reflow • Rework • Quality

Why Soldering Skill Matters

The hardware engineer's most-used physical skill

Prototypes

Hand-soldering closes the idea → working board loop in hours.

Debugging

Lift a pin, add a flying wire, rework a part — all need solder skill.

Rework

Even on production boards, some reworking is unavoidable.

Quality eye

Knowing what a good joint looks like → better designs, better reviews.

Confidence

An engineer who can solder is trusted in any hardware team.

Career

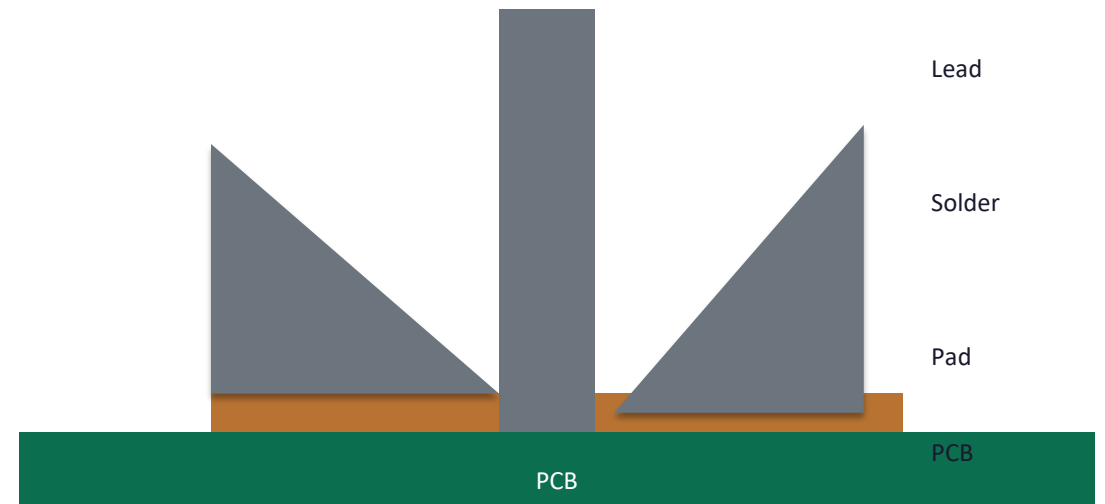
Hardware interviews often include a hands-on soldering task.

How Solder Actually Works

Wetting, intermetallic bond, surface tension

- ▶ Solder is a low-melting-point alloy (Sn-Pb or Sn-Ag-Cu)
- ▶ Heating melts solder → it flows by capillary action
- ▶ Flux strips oxide from copper & solder surface
- ▶ Clean metal + molten alloy → intermetallic compound forms
- ▶ Cooling solidifies the joint — bond strength comes from intermetallic, not solder bulk
- ▶ Without flux, no wetting — the solder just sits in a ball
- ▶ Lead-free (SAC305) melts at $\sim 217^{\circ}\text{C}$; tin-lead (Sn63Pb37) at 183°C

Cross-section of a good solder joint



Good joint: smooth concave fillet, shiny (Sn-Pb) or matte (lead-free).

Solder Types & Compositions

Lead vs lead-free, wire vs paste, fluxes

Type	Composition	Melting	Use	Notes
Sn63Pb37	63% Sn / 37% Pb	183 °C	Hand-soldering (legacy)	Eutectic — sharp melt
Sn60Pb40	60% Sn / 40% Pb	183-188	Common in repair	Slightly pastier transition
SAC305	96.5 Sn / 3 Ag / 0.5 Cu	217-220	Lead-free production	RoHS compliant
SnBi (low-temp)	42 Sn / 58 Bi	138 °C	Heat-sensitive parts	Brittle, low temp
Solder paste (Sn-Pb)	Powder + flux	183 °C	Stencil + reflow (SMT)	Refrigerate
Solder paste (SAC)	Powder + flux	217-225	Lead-free SMT production	RoHS

Flux types

- Rosin (R / RMA) — classic, easy to clean with IPA
- No-clean (NC) — leaves residue (cosmetic) but no need to wash
- Water-soluble (OA) — strong, MUST be washed off

Practical tip

Sn63Pb37 + RMA flux is the easiest combination to learn. Switch to lead-free only when shipping a commercial product (RoHS).

Workspace Setup

The tools every electronics bench needs

Soldering iron / station

Temperature-controlled, ≥ 50 W,
350 °C tip max.

Tip selection

Chisel for through-hole,
conical for fine SMD.

Flux pen / paste

RMA or no-clean flux — essential
for SMD & rework.

Wick + solder sucker

Desoldering braid + pump
for cleanup & rework.

Helping hands / vise

Hold the board, free your hands.
Microscope optional but great.

PPE & ventilation

Safety glasses, fume extractor
(rosin smoke is harmful).

Hand Soldering — The 5-Second Method

Five steps every engineer should be able to demo

1. Apply flux

A drop of flux on the joint before you start. Most beginners skip this — bad joints follow.

2. Heat the joint, not the solder

Touch the iron to BOTH the pad and the lead simultaneously.

3. Feed solder TO the joint

Touch wire to the joint, not the iron. Wait for it to flow.

4. Remove solder first, then iron

Pull solder away ~0.5 s before iron — leaves clean fillet.

5. Inspect

Smooth, concave, shiny/matte. Re-heat with flux if it looks bad.

Anatomy of Good vs Bad Joints

Inspect every joint — train your eye

Good joint

Concave, smooth, shiny (lead) or matte (lead-free). Wets BOTH pad & lead.

Cold joint

Dull, grainy, often peaked. Wetted only one surface. Will fail later.

Insufficient

Too little solder — pad barely covered, lead exposed. Weak mechanically.

Excess solder

Ball or blob on top — could short adjacent pads, hides what's underneath.

Bridge

Solder spans two adjacent pads → short circuit. Common on SOIC / fine pitch.

Tombstone

SMD component stands on one end — one pad heated before the other.

SMD Passive Soldering

0805 / 0603 / 0402 — easier than they look

Method: Tin-and-place

- ▶ Tin ONE pad first with a tiny amount of solder
- ▶ Apply flux to both pads
- ▶ Hold the component with tweezers, reheat tinned pad
- ▶ Slide the component onto the tinned pad
- ▶ Solder the other pad while the first joint sets
- ▶ Reflow the first pad if it looks suspicious

Method: Drag soldering (for ICs)

- ▶ Tack two opposite-corner pins (alignment first!)
- ▶ Apply flux liberally across the row of pins
- ▶ Use a chisel tip with a tiny solder bead
- ▶ Drag slowly across the pins — flux pulls solder onto each
- ▶ Inspect for bridges — wick them off if present
- ▶ Clean with IPA after

Hot Air Rework

For QFNs, BGAs and any leadless component

- ▶ Air temp 280–360 °C, airflow medium
- ▶ Pre-heat the board (hotplate / preheater) — reduces thermal shock
- ▶ Apply flux + a tiny solder bead per pad if needed
- ▶ Place component, hold steady while heating
- ▶ Watch for the 'pop' as solder melts — component self-aligns
- ▶ Don't move — let cool naturally for 30 s
- ▶ Inspect under microscope; reflow joints that look dull

When hot air is needed

- ▶ QFN / DFN — pads under the package, iron can't reach
- ▶ BGA — solder balls, only reflow works
- ▶ Removing failed parts — heat the whole package, lift gently
- ▶ Removing connectors with many pins
- ▶ Re-attaching pads that lifted from old joints
- ▶ Heat-shrinking and assembly tasks

Solder Paste & Stencils

How to get paste onto pads accurately

Steel stencil

Laser-cut from 0.1 mm steel.
Fits over PCB on registration pins.

Apply paste

Bead of paste at top edge,
spread with squeegee at 45°.

Lift stencil

Even paste deposits remain.
Inspect for skips or smears.

Storage

Refrigerate paste (4 °C).
Let warm to room temp before use.

Aperture design

Slightly smaller than pad for fine pitch.
Reduces bridging.

Suppliers

JLCPCB, OSHStencils,
PCBWay — under ₹1500 for prototype.

Common Reflow Defects

What went wrong, and how to fix it

Defect	Cause	Fix
Tombstoning	One pad heated before other	Symmetric pad sizes / pre-heat board
Solder balls	Too much paste; flux residue	Reduce aperture; clean paste
Cold joints	Profile didn't reach peak temp	Higher peak or longer TAL
Bridges	Excess paste; pads too close	Smaller apertures; better stencil
Tarnished joints	Old paste; humid storage	Fresh paste; sealed storage
Lifted pads	Excessive heat or rework	Lower peak; use hot air, not iron
Cracked components	Thermal shock from rapid cool	Slower cool zone profile
Skipped paste	Stencil cleanliness; warped board	Clean stencil between prints

Rework — Removing & Replacing Parts

Every prototype needs rework — make peace with it

- ▶ Identify part type — leaded vs leadless determines method
- ▶ Leaded (SOIC, QFP): apply flux, hot air, lift with tweezers
- ▶ Leadless (QFN, BGA): hot air across the whole package, lift
- ▶ After removal: wick off remaining solder, clean with flux + IPA
- ▶ Re-tin pads lightly before placing new part
- ▶ Lifted pads? Scrape solder mask, use kynar wire to bridge
- ▶ Use Kapton tape to protect adjacent parts from hot air
- ▶ Test electrically AFTER reflow; visually under microscope

Pre-heat the board!

Local hot-air on a cold board cracks substrate, lifts pads, and warps the PCB. ALWAYS preheat to ~100 °C before any rework on a populated board.

Rework rule of thumb

If you've reworked a joint three times, the pad is compromised — move to a different spot (use a kynar wire to the next via) or scrap the board.

Inspection — IPC-A-610 Standards

How industry judges joint quality

- ▶ IPC-A-610 defines Class 1 / 2 / 3 acceptability
- ▶ Class 1 — Consumer (TVs, toys) — most permissive
- ▶ Class 2 — Industrial (servers, telecom) — common standard
- ▶ Class 3 — High-reliability (medical, aerospace) — strictest
- ▶ Wetting angle $\leq 90^\circ$ (joint flows out, not balls up)
- ▶ Solder must cover $\geq 75\%$ of pad on Class 2
- ▶ No bridges, no lifted leads, no fractures
- ▶ Visual + microscope inspection; X-ray for BGA

Useful inspection tools

- ▶ 10× loupe — daily inspection
- ▶ 30–60× digital microscope (₹2-5k)
- ▶ AOI machines — automated optical inspection (production)
- ▶ X-ray — BGA & hidden joints (₹50k+ machine)
- ▶ Multimeter / oscilloscope — electrical confirmation
- ▶ ICT bed-of-nails — production functional test

Safety & Health

Soldering is fume-and-heat work — protect yourself

Fume extraction

Rosin fumes irritate lungs.
Use a fan + carbon filter or open window.

Safety glasses

Solder can pop unexpectedly.
Protect eyes always.

Lead hygiene

Wash hands after handling.
Never eat at the bench.

Burns

Tip is 350 °C — handle hot iron only by handle.
Return to stand always.

Electrical

Unplug bench when working on AC.
Always meter before touching.

Battery hazards

Li-ion + short = fire.
Mask off batteries during soldering.

Recommended Tools by Budget

What to actually buy in 2026

Budget	Iron / Station	Hot air	Reflow	Other
< ₹3,000	Cheap 60 W iron	—	—	Wick, solder, flux
₹5-10k	Hakko T12 clone / Pinecil	858D	Modified toaster oven	Microscope (USB)
₹15-25k	Hakko FX-888D	858D / 8586D	T-962 / phase-change	Better microscope, lighting
₹40-80k	Hakko FM-203 / JBC	JBC HT420	T-962A / CHIPMASTER	Stencil printer, hotplate
Industry	JBC / Metcal	Pace HRT	Heller / BTU convection	AOI, X-ray, full SMT line

Student recommendation

Pinecil + Hakko clone-tips + 858D hot air + Pinecil holder + a USB microscope ≈ ₹6-8k. Covers everything for years.

Key Takeaways

Key takeaways from this lecture

Flux is non-negotiable

Without it, no wetting, no joint.

Heat the joint, not solder

5-second method, every time.

Train your eye

Inspect every joint — develop the IPC-A-610 instinct.

Pre-heat for rework

Cold board + hot air = cracked PCB.

Stencil for production

Reflow + paste outperforms hand-solder past 0603.

Practice deliberately

Build a \$5 kit, deconstruct it, rebuild.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ IPC-A-610 — Acceptability of Electronic Assemblies (the bible)
- ▶ Pace / JBC / Hakko — manufacturer training videos on YouTube
- ▶ EEVblog — 'Soldering Tutorial' classic 3-part series
- ▶ MikesElectricStuff (YouTube) — repair / rework demonstrations
- ▶ Adafruit / Sparkfun — beginner kits + tutorials
- ▶ Pinecil + Pinetool community — affordable modern tooling
- ▶ JLCPCB / PCBWay stencil services — under ₹1500 for protos
- ▶ Robert Feranec — hot-air & BGA rework masterclasses

Thank you • Build something this week.